

COLLEGE OF POPULATION STUDIES • CHULALONGKORN UNIVERSITY

Population Growth and Sources of Demographic Data

Session 2 • From Censuses to Digital Trace Data

Course 5140602 • Fundamental Demographic Analysis (Revised 2026 Syllabus)

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3-hour session



From Session 1: The Toolkit We Bring Today

- We defined demography and its two branches (formal demography and population studies)
- We practiced counts, proportions, ratios, and rates — the building blocks of every indicator
- We situated demography in a data landscape that spans official statistics and emerging digital sources
- **Today we ask: Population growth rate, where do all these numbers actually come from — and how much should we trust them?**



By the End of This Session, You Will Be Able To...



Apply the balancing equation

Decompose population change into births, deaths, and migration



Compute growth & doubling time

Use exponential growth models



Compare official data sources

Evaluate censuses, CRVS, surveys, and registers on coverage and quality



Assess data quality

Detect age heaping using Whipple's Index



Evaluate digital-trace data

Weigh strengths and biases of mobile, satellite, and social-media data



Today's 3-Hour Structure

-  Recap, objectives & agenda
-  Block 1 — The balancing equation & growth-rate math
-  Block 2 — Official/traditional data sources & quality assessment
-  Break
-  Block 3 — Non-traditional & digital-trace data
-  Synthesis: triangulating data sources
-  Wrap-up, Q&A, practice exercises

BLOCK 1

The Balancing Equation & Growth-Rate Math

Population change decomposed; exponential growth; doubling time — the arithmetic every demographer must be fluent in.



Four Components, One Accounting Identity



Births (B)

Additions from fertility —
measured via CRVS or
surveys



Deaths (D)

Losses from mortality —
measured via CRVS or
verbal autopsy



In-migration (IM)

Additions from people
moving into the area



Out-migration (OM)

Losses from people moving
out of the area



The Demographic Balancing Equation

$$P_2 = P_1 + B - D + IM - OM$$

- P_1, P_2 = population at times 1 and 2
- B = births, D = deaths (natural increase = $B - D$)
- IM = in-migration, OM = out-migration (net migration = $IM - OM$)
- This single identity underlies every population estimate, projection, and intercensal check you will ever perform

Fill the missing numbers

Scenario	(P_1)	Births (B)	Deaths (D)	In-migration (IM)	Out-migration (OM)	(P_2)
1. Low fertility, migration-led growth	500,000	4,500	5,000	12,000	3,000	
2. Population decline	800,000	6,000	9,500		7,000	793,500
3. Natural increase but net out-migration	2,000,000		20,000	8,000	25,000	1,995,000



Exponential Growth & the Rule of 70

$$P_2 = P_1 \cdot e^{rn}$$

- r = annual population growth rate, expressed as a decimal.
- n = number of years.
- $e \approx 2.71828$ is the base of natural logarithms; this formula assumes continuous (exponential) compounding.

$$t_2 \approx \frac{70}{r \times 100}$$

- t_2 = approximate number of years required for a population to double at rate r

A discrete-growth alternative is also commonly used

$$P_2 = P_1(1 + r)^n$$

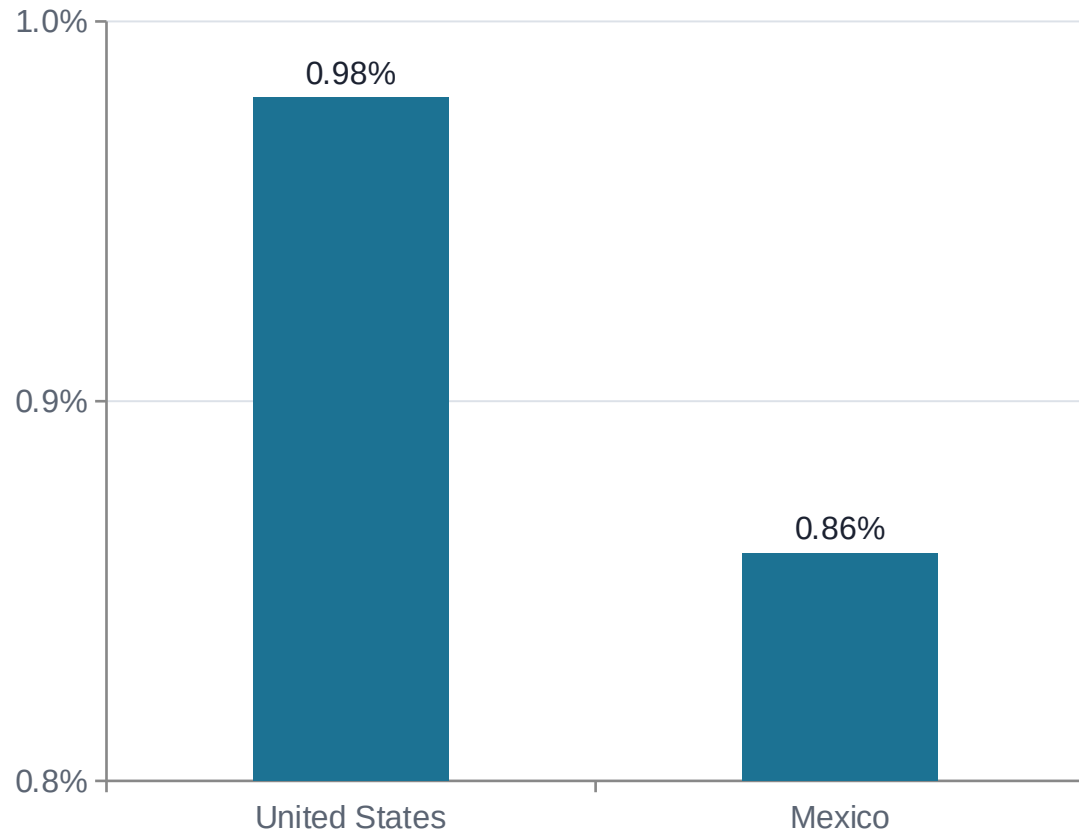
If a population is 10 million and grows continuously at 1% per year for 20 years:

$$P_2 = 10,000,000 \cdot e^{(0.01 \times 20)} \approx 12,214,000$$

Thus, the population would increase to approximately **12.2 million** after 20 years.



Comparing Real Growth Rates: US vs. Mexico, 2024



Using $P_2 = P_1 \cdot e^{rn}$

1. If the U.S. population were 335 million in 2024, with an annual growth rate of $r = 0.0098$, the population in the next 10 year?

it would reach approximately $335 \times e^{(0.0098 \times 10)} \approx 370$ million by 2034, assuming the rate remained constant.

2. Mexico, with $r = 0.0086$, grows more slowly in relative terms despite its younger age structure.



Your Turn: Growth Rate & Doubling Time

Working in pairs (10 minutes):

1. Mexico's population was approximately 130 million in 2024, with an annual growth rate of $r=0.0086$. Using $P_2 = P_1 \cdot e^{rn}$, project Mexico's population in 2034.
2. Suppose Thailand's population declines at a constant annual rate of 0.5%. When will it take for the population to fall to half its current size? What does a negative population growth rate imply about the components of the demographic balancing equation?
3. Explore the current population and annual growth rate of your country. Assuming that the growth rate remains constant, calculate the year in which its population would:
 - increase to 150% of its current size if the growth rate is positive; or
 - decrease to 50% of its current size if the growth rate is negative.

Briefly discuss whether this projection is realistic and identify the assumptions.

BLOCK 2

Official & Traditional Data Sources

The four pillars of official demographic statistics — and how to assess their completeness and quality.



Official Sources of Demographic Data



Census

Complete enumeration of a population at a point in time, typically every 5–10 years



Civil registration & vital statistics **CRVS**

Civil registration & vital statistics — continuous recording of births, deaths, marriages



Sample surveys

Household surveys (e.g., DHS, labour-force surveys) collecting detailed data from a sample



Population registers

Continuously updated administrative records of every resident, common in Nordic countries

Population census

Egyptologists estimate that a large-scale census was taken around 2600 BCE (before Common Era) under the reign of Sneferu (Wilkinson, 2000).

In the 1800s, the practice of taking population censuses spread through Western countries.

After a post-World War II phenomenon – population census has been promoted by the UN.

Source: Jennifer H. Lundquist, Douglas L. Anderton, and Barbara Yaukey (2015) DEMOGRAPHY

Participate in the Population and Housing Census
For a Better Future of Thai Society

ร่วมมือทำสำมะโนประชากรและเคหะ
เพื่อวันพรุ่งนี้ที่ดีกว่าของสังคมไทย



What is population and housing census?
สำมะโนประชากรและเคหะคืออะไร

Population and housing census is a collection of data about every individual residing throughout the country upon which a study will then be conducted to make a comprehensive view in term of numbers of people in each area classified by gender, age, education, nationality, occupation, etc. and housing data such as type of dwelling, tenure of dwelling, source of water for use and drink.

Why should foreign population and employers pay attention and join the population and housing census?

ทำไมประชากรต่างชาติและนายจ้างจึงควรให้ความร่วมมือในการจัดทำสำมะโนประชากร

Data about foreigners in various aspects such as numbers, characteristics, and way of living would be a great benefit for the Government in setting policy, planning and managing matters related to a foreigner in an effective and appropriate manner.

Benefits of population and housing census
ประโยชน์ของการทำสำมะโนประชากร

Generally, the Government would refer to data from population and housing census when setting plan and policy in regard to locality development and improvement to best relieve problems of people in each area of Thailand. The data serves as a helpful resource for making decision in various matters such as management of public utility, development of life quality and environment, management of education, provision of public health service, management of transportation and traffic system.

In this population and housing census, National Statistical Office works jointly with 6 alliances which are Ministry of Interior, Ministry of Public Health, Ministry of Labour, Ministry of Education, Ministry of Social Development and Human Security, Ministry of Agriculture and Cooperatives, Office of National Economic and Social Development Board, and Thailand Post. This cooperation will create a great advantage for development of policy, work plan and projects to effectively help solve problems in each area and maximize the highest benefit to all people.

Trust in the NSO for the Progress of the State and the People.
มั่นใจในสำนักงานสถิติแห่งชาติ รัฐและราษฎรจะก้าวหน้าอย่างมั่นคง



สสช.
สำนักงานสถิติแห่งชาติ

Celebrate the 100th Anniversary of Thailand's Population Census
1-30 September 2010

Please give information to the National Statistical Officials or
Enter the data directly via internet at www.nso.go.th

Type of data collection	Use of registers as census data sources		
	No registers (fully field enumeration based or sample surveys) ²⁵	Base registers (individuals, households, dwellings)	Integrated administrative sources ²⁶
Full field enumeration	Full field enumeration only (traditional census)	Base registers and full field enumeration	Integrated administrative sources and full field enumeration
Rolling surveys (continuous surveys)	Full field enumeration and rolling surveys	n.a.	n.a.
	Rolling census		
Ad hoc sample surveys	n.a.	Base registers and ad hoc sample surveys	Integrated administrative sources and ad hoc sample surveys
Existing sample surveys	n.a.	Base registers and existing sample surveys	Integrated administrative sources and existing sample surveys
No field enumeration (fully register based)	n.a.	n.a.	Fully register based

Basic features of a census

1. Individual enumeration;

The term “census” implies that each individual and each set of living quarters is enumerated separately and that the characteristics thereof are separately recorded.

2. Universality within a defined territory;

Everybody in the specified territory should be similarly counted and described.

3. Simultaneity;

Each person and each set of living quarters should be enumerated as of the same well defined point in time and the data collected should refer to a well-defined reference period.

4. Defined periodicity;

Periodic; Censuses should be taken at regular intervals so that comparable information is made available in a fixed sequence.

5. Capacity to produce small-area statistics;

The census should produce data on the number and characteristics of the population and housing units down to the lowest appropriate geographic level, compatible with national circumstance, and for small population groups, all the while protecting confidentiality of personal information on each individual.

Source: https://unstats.un.org/unsd/demographic-social/Standards-and-Methods/files/Principles_and_Recommendations/Population-and-Housing-Censuses/Series_M67rev3-E.pdf



Official Sources of Demographic Data



Census

Complete enumeration of a population at a point in time, typically every 5–10 years

Advantages

- Provides the most comprehensive picture of the population and households.
- Produces reliable statistics for small geographic areas and population subgroups.
- Establishes benchmark population figures for national and local planning.
- Provides sampling frames and population weights for household surveys.
- Supports the allocation of public resources and political representation.
- Enables comparisons across areas and, when definitions are consistent, over time.

Limitations

- Expensive and operationally complex to conduct.
- Usually conducted only every 5–10 years, so results may become outdated.
- Requires considerable time for data processing, analysis, and publication.
- May undercount hard-to-reach populations, including migrants, homeless people, and residents of remote areas.
- Collects less detailed information than specialised sample surveys.
- Results may be affected by non-response, reporting errors, political sensitivities, or inconsistent definitions.

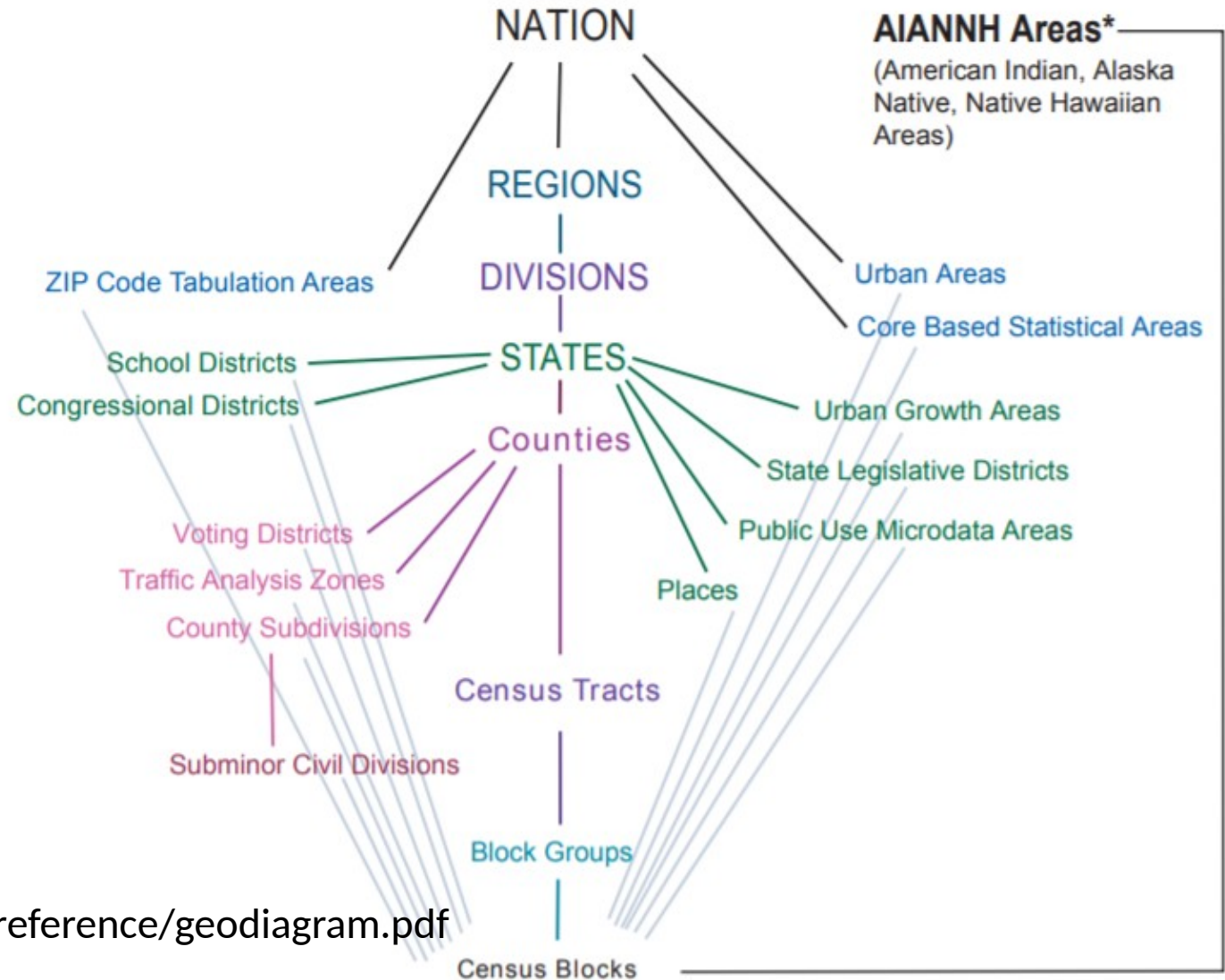


Official Sources of Demographic Data

Modern census

- It specifies the geographic boundaries of the area to be covered.
- The government decides which traits it needs to know about for each of its people and settles upon a questionnaire.
- Simultaneously, within all of the subareas, the government sends out the assigned enumerator to visit households and to list all of the individuals in each household.
- The government compiles the information from the subareas, tabulating numbers of people by subarea (distribution) and by category of individual trait (composition)
- The government reports the findings to the sponsoring citizenry.

Standard Hierarchy of Geographic Entities for the 2010 Decennial Census



Source: <https://www2.census.gov/geo/pdfs/reference/geodiagram.pdf>

What topics to be included?

Political side

The priority of national over international needs in determining census content e.g.

- Hearings and debates dealing with demographic priorities
- Rights to privacy

Technical side

- Complete and comparable.
- Nations can enhance international comparability by taking censuses in standard years

Let's take a look at the Thai census

<https://unstats.un.org/unsd/demographic/sources/census/quest/THA2010en.pdf>

<http://www.nso.go.th/sites/2014en/Pages/popeng/2010/mainpop.aspx>

Source: Jennifer H. Lundquist, Douglas L. Anderton, and Barbara Yaukey (2015)
DEMOGRAPHY

EXAMPLE 1

Thailand's 2025 Census Goes Digital- First



- Thailand's National Statistical Office ran the 2025 Population and Housing Census under a "Digital First Approach," directing respondents to self-report via a web application and government portal before enumerator follow-up
- This is a major shift from the 2020 census round, when online/mailed pilot response rates were only about 5% and 4% respectively
- Censuses are costly (they attempt to count everyone) and infrequent — but they remain the gold-standard sampling frame that surveys and small-area estimates are built on

Source: United Nations ESCAP (2024). Thailand Population and Housing Census 2025 [Conference presentation]. Asia-Pacific Stats Café Series. https://crvs.unescap.org/system/files/sites/default/files/Thailand_presentation.pdf

EXAMPLE 2

Thailand's Near- Complete Civil Registration



- By 2024, Thailand reached essentially 100% registration of births occurring within its territory in that year, as part of the Asia-Pacific CRVS Decade (2015–2024) coordinated by UN ESCAP
- Challenges remain: a high share of deaths occur outside health facilities, requiring special programs to improve medically certified cause-of-death coding
- CRVS data — unlike a census — are continuous, giving up-to-date annual counts of births and deaths, the numerator for CBR, CDR, and related rates

Checkpoint

- Find Population and Housing Census from [United Nations Statistics Division - Demographic and Social Statistics](https://unstats.un.org/unsd/demographic-social/)
- <https://unstats.un.org/unsd/demographic-social/census/document-resources/>

Select two Population and Housing Census questionnaires from two different countries, and provide the criteria for your selection.

You will have 15 minutes to compare the questionnaires between the two countries. Afterward, share 2-3 observations related to the content of the questionnaires. Each group will have 3 minutes to report back.

How to cope with discrepancies?

For example

- <https://www.census.gov/newsroom/press-releases/2022/2020-census-estimates-of-undercount-and-overcount.html>
- <https://www.census.gov/library/stories/2022/05/2020-census-undercount-overcount-rates-by-state.html#:~:text=If%20an%20estimated%20net%20coverage,error%20or%20more%20than%20once>

// Census.gov / America Counts: Stories Behind the Numbers / Census Bureau Today Releases 2020 Census Undercount, Overcount Rates by State



Census Bureau Today Releases 2020 Census Undercount, Overcount Rates by State



2020 Census Undercounts in Six States, Overcounts in Eight





Official Sources of Demographic Data



Civil registration &
vital statistics

CRVS

Civil registration &
vital statistics —
continuous recording
of births, deaths,
marriages

Advantages

- Continuously records vital events, including births, deaths, marriages, divorces, and sometimes causes of death.
- Produces more timely demographic information than a periodic census.
- Provides legal identity and documentation needed to access rights and public services.
- Supports the calculation of fertility, mortality, life expectancy, and cause-of-death indicators.
- Can generate statistics for small geographic areas and monitor changes over time.
- Covers the entire population in principle rather than only a sample.

Limitations

- Registration may be incomplete, particularly in remote areas and among migrants or marginalised populations.
- Some events may be registered late, inaccurately, or more than once.
- Cause-of-death information may be missing or incorrectly classified.
- Data quality and coverage can vary substantially across regions and population groups.
- Civil registration systems may be fragmented across different government agencies.
- Privacy, confidentiality, and restricted data access may limit research use.



Sample Surveys and Population Registers

Sample Surveys

- Collect rich, detailed data (fertility histories, health, income) from a representative sample, not everyone
- Examples: Demographic and Health Surveys (DHS), Thailand's Labour Force Survey (NSO)
- Fast and flexible, but subject to sampling error and potential non-response bias

Population Registers

- Continuously updated administrative databases covering every resident (e.g., Sweden's Register of Total Population)
- Enable register-based "censuses" without door-to-door enumeration
- Require a unique personal ID system and strong inter-agency data-sharing — not feasible everywhere

EXAMPLE 3

Population Registers as a Data Source: Denmark & Sweden



- Careja and Bevelander (2018) show how Danish and Swedish population registers — built on unique personal identity numbers — provide systematic, longitudinal, near-complete data on residents, including immigrants
- Register-linked data let researchers follow the same individuals across health, tax, and migration records over decades, something few surveys can match
- The tradeoff: registers are only as good as administrative coverage and timely updating, and this model depends on high public trust and legal infrastructure most countries do not yet have



Strengths and Limitations at a Glance

Source	Key strength	Key limitation
Census	Universal coverage; small-area detail	Costly; infrequent (5–10 yr gaps)
CRVS	Continuous, real-time vital events	Completeness varies; cause-of-death quality issues
Sample surveys	Rich detail; flexible design	Sampling error; possible non-response bias
Population registers	Continuous, near-complete, linkable	Requires strong ID/legal infrastructure



Your Turn

Working in small groups, select one country and address the following questions:

1. What official demographic data sources are available in your country?

Identify examples of the population census, civil registration and vital statistics (CRVS), sample surveys, and population registers. For each source, provide the responsible institution, latest available year, frequency of data collection, population coverage, and an official link. If a source is unavailable, explain why.

2. How do these sources differ in purpose and data quality?

Compare their major indicators, geographic coverage, timeliness, completeness, accessibility, and possible sources of error or undercoverage. Which population groups are most likely to be omitted or inadequately represented?

3. How should the sources be combined?

Select one demographic issue—such as fertility, mortality, migration, ageing, or population growth—and explain which source would be most appropriate. Discuss how another source could be used to validate or complement the findings.



Why Assess Completeness and Quality?

- Every data source can misstate ages, undercount hard-to-reach groups, or lag behind real events
- Age misreporting is one of the most common and best-studied data-quality problems
 - respondents (or enumerators) round ages to preferred digits, especially 0 and 5
- This "age heaping" distorts age-specific rates, life tables, and population pyramids if left uncorrected
- The UN formalized a simple diagnostic — Whipple's Index — in its 1955 Manual II, still used by statistical agencies today



Whipple's Index: Detecting Age Heaping

$$WI = [5 \times \sum P(25,30,...,60) / \sum P(23...62)] \times 100$$

- Numerator: sum of persons reporting ages ending in 0 or 5, from 25 to 60
- Denominator: total persons aged 23–62 inclusive
- $WI = 100 \rightarrow$ no age preference; $WI = 500$
- $\rightarrow$ everyone reports a age ending 0/5
- UN benchmark: <105 "highly accurate", $105-109.9$ "fairly accurate" ... >175 "very rough" data

Suppose a household survey of 10,000 people aged 23–62 reports:

- • 2,300 people report ages ending in 0 or 5 (e.g., 25, 30, 35 ... 60)
- • Total surveyed aged 23–62 = 10,000

$$WI = (5 \times 2,300 / 10,000) \times 100 = 115$$

Interpretation: $WI = 115$ falls in the UN's "fairly accurate" to "approximate" range — noticeable preference for ages ending in 0 or 5, warranting caution in single-year age analysis.

LET'S DISCUSS

Why Does Age Heaping Happen?



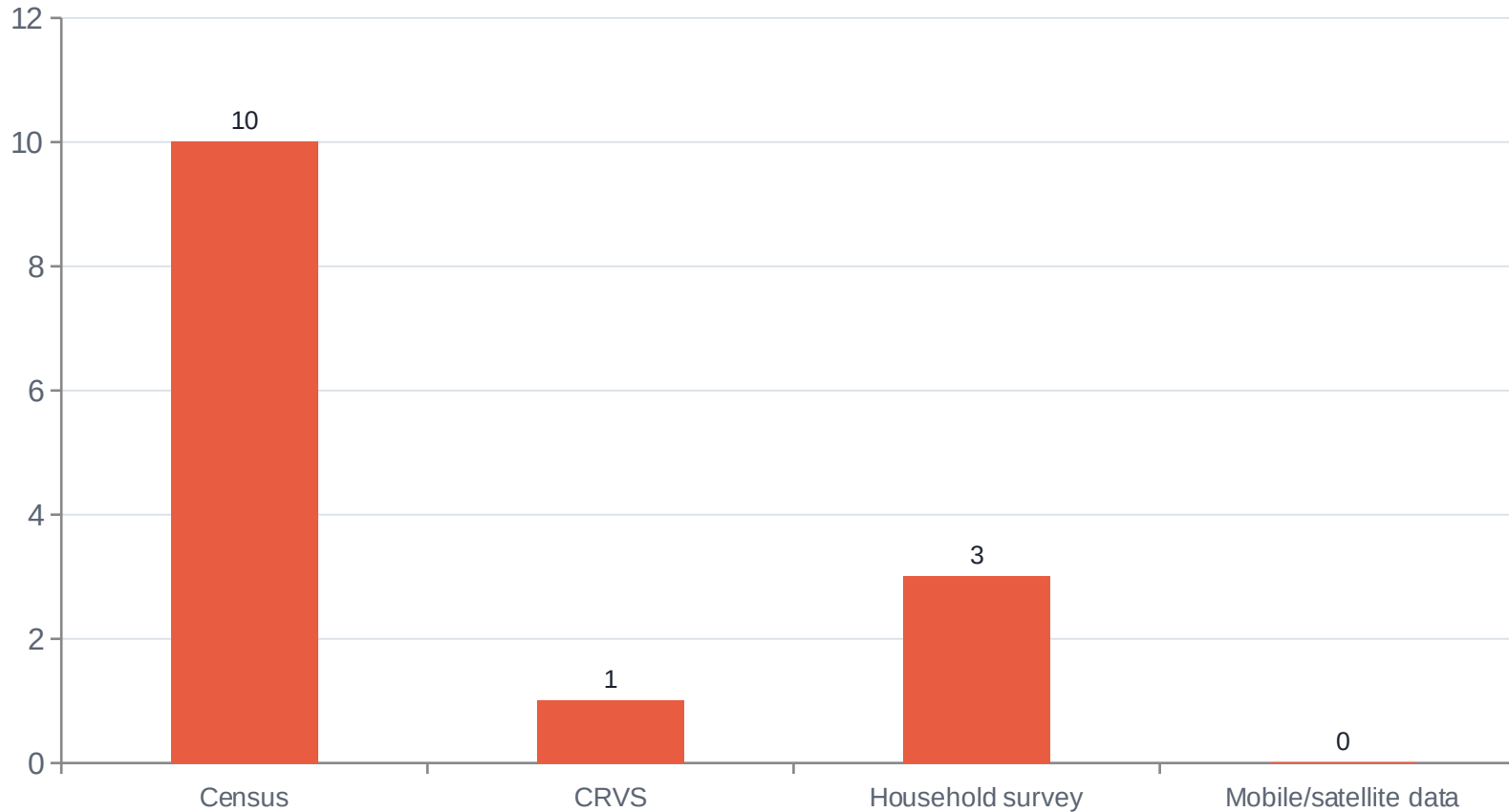
- What individual-level factors (education, literacy, birth documentation) might drive age heaping in a household survey?
- Why might age heaping be worse in older cohorts than younger ones in the same country?
- How might a National Statistical Office correct age-specific rates once it detects heaping?

BLOCK

Non-Traditional & Digital-Trace Data

Mobile-phone records, satellite imagery, and social-media data as fast, high-resolution complements to official statistics — with real caveats.

Official Data Update Slowly — Digital Signals Update Fast



Illustrative update latency (years) for population information, by source type.

Censuses may be a decade old by the time of use; even annual CRVS/survey data lag real-time events such as displacement or rapid urban growth.

Digital-trace data can update daily — motivating its use as a complement, not a replacement, for official statistics.



Mobile-Phone Call Detail Records (CDRs)

- CDRs are logs mobile operators generate for billing: which cell tower a call/text connected through, and when
- Aggregated and anonymized, tower-level CDRs reveal where subscribers are, in near-real time, across an entire country
- Because CDRs are collected passively for billing (not demographic) purposes, they can achieve national coverage at a fraction of a census's cost
- Key limitation: only captures mobile subscribers — coverage gaps for children, the very poor, and multiple-SIM users must be modeled or corrected

EXAMPLE 4

Dynamic Population Mapping From Mobile Data



- Deville et al. (2014, PNAS) used mobile-operator call data in Portugal and France to produce dynamic, fine-grained maps of population distribution — including daily commuting patterns official censuses cannot capture
- The method estimates population density by combining call-volume patterns with a "top-down" census-based prior, so it stays anchored to official totals while adding spatial and temporal detail
- The authors explicitly note applicability to low-income countries where census data are "scarce, outdated, or unreliable" — precisely the settings where traditional data are weakest



Satellite Imagery-Based Population Mapping

- Satellite imagery detects built-up areas, night-lights, and settlement footprints at very high spatial resolution (down to ~100m grid cells)
- WorldPop (University of Southampton) combines satellite/geospatial layers with census counts and, increasingly, mobile-phone data using machine-learning (random-forest) models to redistribute population within census units
- This produces gridded population estimates useful for disaster response, vaccination-campaign planning, and small-area estimation between censuses
- Limitation: it models where structures are, not who lives in them — demographic composition (age, sex) still typically comes from census/survey data

EXAMPLE 5

WorldPop: Bottom-Up, Open Population Grids



- WorldPop produces open-access, gridded population datasets for most countries, rebuilding population distribution from satellite-derived settlement layers calibrated against census totals
- Its "bottom-up" projects have directly supported humanitarian response, malaria-elimination planning, and vaccination micro-planning in settings with outdated census data
- For Thailand and the wider region, WorldPop-style products let researchers estimate population in areas or years the last census did not directly cover — a natural bridge to the mobile and satellite themes you'll revisit in the Applied Classes on AI/digital data



Social Media & Internet Data as Demographic Signal

- Platforms like Facebook's advertising audience tool or search/e-mail data can proxy migration stocks, sentiment, or even mortality shocks in near-real time
- Zagheni and Weber (2015) show these sources are cheap and timely, but fundamentally non-representative: internet/social-media users differ systematically from the general population by age, income, gender, and geography
- Bias is not fixed — it varies by platform, country, and over time, so naive extrapolation from "who posts" to "who exists" can badly mislead policy
- The responsible approach: calibrate digital signals against a known benchmark (census or survey) rather than using them as stand-alone population counts

EXAMPLE 6

The Representativeness Problem, Named



- Zagheni & Weber (2015) formalize why "big" internet data is not automatically "representative" data: coverage is skewed toward younger, urban, wealthier, and more digitally connected populations
- Their earlier e-mail-based migration-rate study illustrates the promise (fast, low-cost signals of international mobility) alongside the same caveat — corrections for selection bias are essential before treating results as population estimates
- This is the central methodological lesson for the whole "digital demography" --- verify before you trust



IPUMS International: Open Census Microdata for Research

- **IPUMS International (Minnesota Population Center, University of Minnesota) harmonizes and disseminates census microdata samples from over 100 countries, including Thailand, spanning 1960s–present**
- Variables are standardized across countries and time so researchers can build genuinely comparative demographic analyses without re-cleaning raw national data
- Access is free to qualified researchers under a usage agreement — a strong complement to the proprietary/commercial digital-data sources discussed above
- A good habit for your applied research paper: check IPUMS International before assuming microdata for a comparative question does not exist

EXAMPLE 7

Using IPUMS International in Practice



- IPUMS International currently distributes samples from 82+ countries, 277+ censuses, and over 600 million individual person records — one of the largest harmonized microdata resources in the social sciences
- A typical project: comparing household size or female labour-force participation across Southeast Asian censuses using consistent variable definitions
- Because it draws on national census microdata, IPUMS International sits at the intersection of "traditional" (census-based) and "modern" (globally harmonized, programmatically accessible) demographic data infrastructure

Traditional vs. Digital-Trace Data: A Balanced View



Dimension	Traditional (census/CRVS/survey)	Digital-trace (mobile/satellite/social)
Coverage	Near-universal by design	Passive; depends on device/platform use
Timeliness	Slow (years)	Near real-time (days/hours)
Representativeness	High, if well-executed	Often biased toward connected sub-groups
Cost	High per data point	Low marginal cost once infrastructure exists
Best use	Benchmark totals & structure	Fine-grained, fast-moving detail — calibrated to a benchmark

LET'S DISCUSS

Triangulating Traditional and Digital Data



- For a rapid-onset disaster in a data-poor region, which combination of sources would you reach for first, and why?
- If a mobile-network operator offered you CDR data for a Thai province, what would you need from NSO to calibrate it responsibly?
- Where do you expect this "traditional + digital" theme to reappear later in the course (hint: Applied Classes 11–13)?



Key Takeaways From Session 2



One accounting identity

$P_2 = P_1 + B - D + IM - OM$ underlies every population estimate



Growth is exponential

$P_2 = P_1 \cdot e^{(rn)}$ and the Rule of 70 let you project and compare growth quickly



Four official pillars

Census, CRVS, surveys, and registers each trade off cost, timeliness, and coverage



Always check quality

Whipple's Index and similar tools reveal age heaping and other reporting errors



Digital data complements

Mobile, satellite, and social data add speed and resolution — but need calibration against a benchmark



Use open resources

IPUMS International and WorldPop give you free, research-grade data for your applied paper

QUESTIONS?

Next session: Mortality & Morbidity

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PRACTICE EXERCISES

14 questions — conceptual & computational — to consolidate Session 2



The Balancing Equation & Growth (1–5); Official Data Sources (6–9)

- 1. Write the demographic balancing equation and define each term.
- 2. A city had 500,000 residents on Jan 1. During the year: 8,000 births, 4,500 deaths, 6,000 in-migrants, 3,000 out-migrants. Find the population on Dec 31.
- 3. A country grows at $r = 0.015$ (1.5%) per year. Using $P_2 = P_1 \cdot e^{(rn)}$, project its population in 20 years if $P_1 = 40,000,000$.
- 4. Using the Rule of 70, estimate the doubling time for a population growing at 3.5% per year.
- 5. Explain why a negative net migration could offset positive natural increase, and give a real-world example.
- 6. Compare a census and a population register on timeliness, coverage, and cost.
- 7. Why might CRVS data understate infant deaths in a country with many home births and deaths?
- 8. Give one advantage sample surveys have over censuses, and one advantage censuses have over surveys.
- 9. Explain in your own words why Thailand's move to a "Digital First" census approach might improve OR threaten data completeness — argue both sides.



Data Quality: Whipple's Index (10–11); Non-Traditional & Digital Data (12–14)

- 10. A sample of 5,000 people aged 23–62 shows 1,400 reporting ages ending in 0 or 5. Calculate Whipple's Index and classify the data quality using the UN benchmark scale.
- 11. Suggest two reasons an enumerator (rather than the respondent) might introduce age heaping into census data.
- 12. Explain how Deville et al. (2014) used mobile-phone call detail records to map population, and name one limitation of this method.
- 13. What does it mean to say social-media data is "non-representative," and how can researchers correct for this (per Zagheni & Weber, 2015)?
- 14. Design a one-paragraph research plan that combines a traditional source (census, CRVS, or survey) with a digital-trace source (mobile, satellite, or social-media data) to answer a real population question of your choice.

THANK YOU

Session 3 — Mortality, Morbidity, and Health

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